* Question 1 $\Rightarrow$

$1 \Rightarrow$

$J = 9-1 = 8 = I$

$S = 18-1 = 17 = R$

$V = 21-1 = 20 = U$

$Y = 24-1 = 23 = X$

$Q = 16-1 = 15 = P$

$2 \Rightarrow$

$J = 9-2 = 7 = H$

$S = 18-2 = 16 = Q$

$V = 12-2 = 19 = T$

$Y = 24-2 = 22 = W$

$Q = 16-2 = 14 = O$

$3 \Rightarrow$

$J = 9-3 = 6 = G$

$S = 18-3 = 15 = P$

$V = 21-3 = 18 = S$

$Y = 24-3 = 21 = V$

$Q = 16-3 = 13 = N$

$4 \Rightarrow$

$J = 9-4 = 5 = F$

$S = 18-4 = 14 = O$

$V = 21-4 = 17 = R$

$Y = 24-4 = 20 = U$

$Q = 16-4 = 12 = M$

$\therefore$ with $k=4$, $\Rightarrow$ FORUM.

Question 2 ⇒

$y = 15x + 19$

$x = y - 19 / 15$

$$C = E(P) = 15p + 19 \pmod{26}$$

$$P = D(C) = 15^{-1}(C - 19) \pmod{26}$$

⇒ (15, 26) ⇒

	15	26
$26 - 15 \times 1 = 11$	$0 - 1 = -1$	$1 - 0 = 1$
$15 - 11 \times 1 = 4$	$1 - -1 = 2$	$0 - 1 = -1$
$11 - 2 \times 4 = 3$	$-1 - 2 \cdot 2 = -5$	$1 + 2 = 3$
$4 - 3 \times 1 = 1$	$2 - -5 = 7$	$-1 - 3 = -4$

$$1 = 7 \times 15 - 4 \times 26$$

$$\therefore 15^{-1} \equiv 7 \pmod{26}$$

$$\therefore p = 7(C - 19) \pmod{26}$$

Q	O	J	B	G	M	D	U	J	K	
16	14	9	1	6	12	3	20	9	10	c
$7 \cdot -3$	$7 \cdot -5$	$7 \cdot -10$	$7(-18)$	$7(-13)$	$7(-7)$	$7(-16)$	$7(1)$	$7(-10)$	$7(-9)$	$7(c-19)$
-21	-35	-70	-126	-91	-49	-112	7	-70	-63	
5	17	8	4	13	3	18	7	8	15	m26
F	R	I	E	N	D	S	H	I	P	

* Question 3 $\Rightarrow$ Y C | N Q | H D , $M = \begin{bmatrix} 20 & 5 \\ 11 & 25 \end{bmatrix}$

for $M^{-1} \Rightarrow$ $\det(M) = 20 \cdot 25 - 5 \cdot 11 = 445 \ m26 = 3$

in a question from the learning evidence,
we found that $3^{-1} \equiv 9 \pmod{26}$

$$M^{-1} = (\det M)^{-1} [M]^{T} \pmod{26}$$

$$= 9 \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$$= 9 \times \begin{bmatrix} 25 & -5 \\ -11 & 20 \end{bmatrix} \pmod{26}$$

$$= 9 \times \begin{bmatrix} 25 & 21 \\ 15 & 20 \end{bmatrix} = \begin{bmatrix} 225 & 189 \\ 135 & 180 \end{bmatrix} \pmod{26}$$

$$M^{-1} = \begin{bmatrix} 17 & 7 \\ 5 & 24 \end{bmatrix}$$

* YC $\Rightarrow$

$$\begin{bmatrix} 24 & 2 \end{bmatrix} \begin{bmatrix} 17 & 7 \\ 5 & 24 \end{bmatrix} \begin{bmatrix} 2 & 8 \end{bmatrix}$$

$24 \cdot 17 + 2 \cdot 5 = 418 \ m26 = 2$

$24 \cdot 7 + 2 \cdot 24 = 216 \ m26 = 8$

YC $\Rightarrow$ CI

* NQ $\Rightarrow$

$$\begin{bmatrix} 13 & 16 \end{bmatrix} \begin{bmatrix} 17 & 7 \\ 5 & 24 \end{bmatrix} \begin{bmatrix} 15 & 7 \end{bmatrix}$$

$13 \cdot 17 + 16 \cdot 5 = 301 \ m26 = 15$

$13 \cdot 7 + 16 \cdot 24 = 475 \ m26 = 7$

$\therefore$ NQ $\Rightarrow$ PH

$HD \Rightarrow$

$$[7 \quad 3]\begin{bmatrix} 17 & 7 \\ 5 & 24 \end{bmatrix} [4 \quad 17]$$

$7 \cdot 17 + 3 \cdot 5 = 134 \bmod 26 = 4$

$7 \cdot 7 + 3 \cdot 24 = 121 \bmod 26 = 17$

$\therefore HD \Rightarrow ER$

$\therefore$ YCNQHD $\Rightarrow$ CIPHER.

* Question 4 $\Rightarrow$ 2x2 matrix

A	P	P	E	A	L	Y	H	B	R	C	T
0	15	15	4	0	11	24	7	1	17	2	19

$$P = \begin{bmatrix} 0 & 15 \\ 15 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 24 & 7 \\ 1 & 17 \end{bmatrix}$$

$$PM = C$$

$$\times P^{-1} \Rightarrow \quad P \times P^{-1} \times M = P^{-1} C$$

$$M = P^{-1} C$$

$P^{-1} \Rightarrow \quad \det(P) = 0 \cdot 4 - 15 \cdot 15 = -225 \ m\,26 = 9 \ (\bmod 26)$

$9^{-1} \Rightarrow (9, 26) \Rightarrow$

	$\underline{\underline{9}}$	$\underline{\underline{26}}$
$26 - 9 \times 2 = 8$	$0 - 2 = -2$	$1 - 0 = 1$
$9 - 8 \times 1 = 1$	$1 + 2 = 3$	$0 - 1 = -1$

$$1 = 3 \times 9 - 26 \;//$$

$$\therefore 9^{-1} \equiv 3 \ (\bmod 26)$$

$$P^{-1} = (\det P)^{-1} \times \operatorname{adj}(P) \quad (\bmod 26)$$

$$= 3 \times \begin{bmatrix} 0 & 15 \\ 15 & 4 \end{bmatrix}$$

$$= 3 \times \begin{bmatrix} 4 & -15 \\ -15 & 0 \end{bmatrix} = 3 \times \begin{bmatrix} 4 & 11 \\ 11 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & 33 \\ 33 & 0 \end{bmatrix} \ (\bmod 26)$$

$$P^{-1} = \begin{bmatrix} 12 & 7 \\ 7 & 0 \end{bmatrix} \pmod{26}$$

$$\Rightarrow M = P^{-1} C \pmod{26}$$

$$= \begin{bmatrix} 12 & 7 \\ 7 & 0 \end{bmatrix} \begin{bmatrix} 24 & 7 \\ 1 & 17 \end{bmatrix} \pmod{26}$$

$$\begin{array}{cc} & \begin{bmatrix} 12 & 7 \\ 7 & 0 \end{bmatrix} \\ \begin{bmatrix} 24 & 7 \\ 1 & 17 \end{bmatrix} & \begin{bmatrix} 9 & 21 \\ 12 & 23 \end{bmatrix} \end{array}$$

$12 \cdot 24 + 7 \cdot 1 = 295 \ m26 = 9$

$12 \cdot 7 + 7 \cdot 11 = 203 \ m26 = 12$

$7 \cdot 24 + 0 = 168 \ m26 = 12$

$7 \cdot 7 + 0 = 49 \ m26 = 23$

$$M = \begin{bmatrix} 9 & 21 \\ 12 & 23 \end{bmatrix} \pmod{26}$$

* Results $\Rightarrow$

classical-exam

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	
Question 1	1 / 1	Review
Question 2	1 / 1	Review
Question 3	1 / 1	Review
Question 4	1 / 1	Review
Total	**4 / 4 (100%)**	

Performance Summary

Exam Name:	classical-exam
Session ID:	16152647147
Exam Start:	Tue Aug 04 2026 20:20:09
Exam Stop:	Tue Aug 04 2026 21:23:56
Time Spent:	1:03:46

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